

Supplementary Material S4: An example workflow

This section demonstrates an example workflow of MinNet (AMCP proposed in this study) on a mineral compositional dataset comprised of 33 unknown mineral compositions.

Input data: The input data (Figure 1) should be in excel spreadsheet. Please note the the oxide names are case-sensitive and should not contain leading or trailing spaces (although order of oxides do not matter). The first column should always be present and named ‘Sample’.

	A	B	C	D	E	F	G	H	I	J	K
1	Sample	SiO2	MgO	FeO	TiO2	Al2O3	MnO	CaO	Na2O	K2O	Total
2	1	50.2	12.8	0.84	0	33.5	0.06	0.23	0.26	0.14	98.03
3	2	47.69	7.56	8.04	0	32.52	0.04	0.52	0.53	0.42	97.37
4	3	44.64	3.08	11.02	0	29.96	1.86	0.02	2.05	0.75	93.38
5	4	41.33	19.6	10.44	0.28	21.83	0.44	4.4	0	0	98.32
6	5	36.7	0.9	29.9	0.75	21.4	1.14	9.02	0	0	99.81
7	6	39.04	0.73	5	0.12	20.43	0.34	34.29	0	0	99.95
8	7	45.24	0.08	0.02	0.01	36.85	0.12	0	0.64	10.08	93.04
9	8	45.45	0.69	1	0.48	35.99	0	0	0.65	10.79	95.05
10	9	50.72	2.37	3.7	0.49	27.65	0.03	0	0.17	11.53	96.66
11	10	27.22	2.34	13.78	0.56	54.16	0.23	0	0	0	98.29
12	11	26.62	5.31	9.75	1.67	55.15	0.03	0	0	0	98.53
13	12	50.2	12.8	0.84	0	33.5	0.06	0.23	0.26	0.14	98.03
14	13	47.69	7.56	8.04	0	32.52	0.04	0.52	0.53	0.42	97.37
15	14	44.64	3.08	11.02	0	29.96	1.86	0.02	2.05	0.75	93.38
16	15	41.33	19.6	10.44	0.28	21.83	0.44	4.4	0	0	98.32
17	16	36.7	0.9	29.9	0.75	21.4	1.14	9.02	0	0	99.81
18	17	39.04	0.73	5	0.12	20.43	0.34	34.29	0	0	99.95
19	18	45.24	0.08	0.02	0.01	36.85	0.12	0	0.64	10.08	93.04
20	19	45.45	0.69	1	0.48	35.99	0	0	0.65	10.79	95.05
21	20	50.72	2.37	3.7	0.49	27.65	0.03	0	0.17	11.53	96.66
22	21	27.22	2.34	13.78	0.56	54.16	0.23	0	0	0	98.29
23	22	26.62	5.31	9.75	1.67	55.15	0.03	0	0	0	98.53
24	23	50.2	12.8	0.84	0	33.5	0.06	0.23	0.26	0.14	98.03
25	24	47.69	7.56	8.04	0	32.52	0.04	0.52	0.53	0.42	97.37
26	25	44.64	3.08	11.02	0	29.96	1.86	0.02	2.05	0.75	93.38
27	26	41.33	19.6	10.44	0.28	21.83	0.44	4.4	0	0	98.32
28	27	36.7	0.9	29.9	0.75	21.4	1.14	9.02	0	0	99.81
29	28	39.04	0.73	5	0.12	20.43	0.34	34.29	0	0	99.95
30	29	45.24	0.08	0.02	0.01	36.85	0.12	0	0.64	10.08	93.04
31	30	45.45	0.69	1	0.48	35.99	0	0	0.65	10.79	95.05
32	31	50.72	2.37	3.7	0.49	27.65	0.03	0	0.17	11.53	96.66
33	32	27.22	2.34	13.78	0.56	54.16	0.23	0	0	0	98.29
34	33	26.62	5.31	9.75	1.67	55.15	0.03	0	0	0	98.53

Figure 1: Screenshot showing an example of input data in Microsoft Excel™ spreadsheet.

User-interface: The initial web-app interface (Figure 2) is composed of three sections.

MinNet

(automated mineral classification program)

Glossary

Oxide-Sum threshold refers to minimum oxide sum below which classifier will consider the analysis wrong and won't classify it.

Minimum Prediction threshold confidence below which classifier thinks its prediction is incorrect and will not report it

Oxide Sum Threshold [Values between 0 and 100]

90.0

Predictive confidence Threshold [Values between 0 and 1]

0.9

Choose a file

 Drag and drop file here
Limit 200MB per file

Browse files

Figure 2: Screenshot of web-app showing user interface. The ‘Drag and Drop’ area and ‘Browse Files’ button is highlighted with red rectangle.

First and second sections are used for defining and setting values for “Oxide-sum threshold” and “Minimum Prediction threshold”. Oxide-sum threshold (defaults to 90) refers to the minimum oxide sum below which mineral analyses are not fed to the classifier whereas Minimum Prediction Threshold (defaults to 0.9) refers to minimum prediction confidence below which classifier will not report the classification. Mineral analyses falling below both the threshold will be marked as “??”.

Third section (highlighted by red rectangle in Figure 2) is used for uploading the mineral composition dataset (in excel format). The input data (in excel workbook format shown in Figure 1) can be uploaded by ‘drag and drop’ method or clicking on the ‘Browse files’ button in the web-app interface.

Result: Upon successful upload of the input data, web-app automatically runs the classification routine and two more sections appear,

- (1) “Classification Summary” (in Figure 3) presents a brief summary of mineral instances correctly identified is given in. Minerals with oxide-sum or prediction confidence below user-set “Oxide-Sum Threshold” and “Prediction confidence threshold” respectively, are reported as “??”.

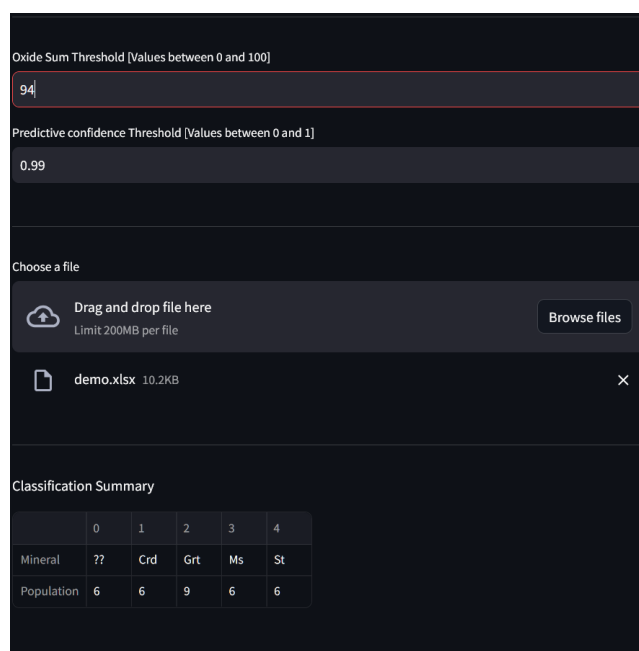


Figure 3: Screenshot of MinNet user interface showing Classification summary of the input data.

- (2) A detailed table (Figure 4) reporting each instance of classified minerals and their calculated mineral formulas. Minerals marked with “??” and 100% confidence means the

mineral was not run due to “Oxide Sum Threshold” and mineral formula is not calculated for mineral marked with “?”. The table can be resized according to user’s need.

Sample	SiO2	MgO	FeO	TiO2	Al2O3	MnO	CaO	Na2O	K2O	Total	Mineral	Confidence
1	50.2	12.8	0.84	0	33.5	0.06	0.23	0.26	0.14	98.03	Crd	100
2	47.69	7.56	8.04	0	32.52	0.04	0.52	0.53	0.42	97.32	Crd	100
3	44.64	3.08	11.02	0	29.96	1.86	0.02	2.05	0.75	93.38	??	100
4	41.33	19.6	10.44	0.28	21.83	0.44	4.4	0	0	98.32	Grt	100
5	36.7	0.9	29.9	0.75	21.4	1.14	9.02	0	0	99.81	Grt	100
6	39.04	0.73	5	0.12	20.43	0.34	34.29	0	0	99.95	Grt	100
7	45.24	0.08	0.02	0.01	36.85	0.12	0	0.64	10.08	93.04	??	100
8	45.45	0.69	1	0.48	35.99	0	0	0.65	10.79	95.05	Ms	100
9	50.72	2.37	3.7	0.49	27.65	0.03	0	0.17	11.53	96.66	Ms	100
10	27.22	2.34	13.78	0.56	54.16	0.23	0	0	0	98.29	St	100
11	26.62	5.31	9.75	1.67	55.15	0.03	0	0	0	98.53	St	100
12	50.2	12.8	0.84	0	33.5	0.06	0.23	0.26	0.14	98.03	Crd	100
13	47.69	7.56	8.04	0	32.52	0.04	0.52	0.53	0.42	97.32	Crd	100

Sample	SiO2	MgO	FeO	TiO2	Al2O3	MnO	CaO	Na2O	K2O	Total	Mineral	Confidence
1	50.2	12.8	0.84	0	33.5	0.06	0.23	0.26	0.14	98.03	Crd	100
2	47.69	7.56	8.04	0	32.52	0.04	0.52	0.53	0.42	97.32	Crd	100
3	44.64	3.08	11.02	0	29.96	1.86	0.02	2.05	0.75	93.38	Crd	100
4	41.33	19.6	10.44	0.28	21.83	0.44	4.4	0	0	98.32	Grt	100
5	36.7	0.9	29.9	0.75	21.4	1.14	9.02	0	0	99.81	Grt	100
6	39.04	0.73	5	0.12	20.43	0.34	34.29	0	0	99.95	Grt	100
7	45.24	0.08	0.02	0.01	36.85	0.12	0	0.64	10.08	93.04	Ms	100
8	45.45	0.69	1	0.48	35.99	0	0	0.65	10.79	95.05	Ms	100
9	50.72	2.37	3.7	0.49	27.65	0.03	0	0.17	11.53	96.66	Ms	100
10	27.22	2.34	13.78	0.56	54.16	0.23	0	0	0	98.29	St	100
11	26.62	5.31	9.75	1.67	55.15	0.03	0	0	0	98.53	St	100
12	50.2	12.8	0.84	0	33.5	0.06	0.23	0.26	0.14	98.03	Crd	100
13	47.69	7.56	8.04	0	32.52	0.04	0.52	0.53	0.42	97.32	Crd	100

Figure 4: Screenshot of MinNet user interface showing detailed table of classified input data (a) with oxide-sum set to 94 and (b) with oxide-sum threshold set to 90. Two instances (Muscovite and cordierite) were not run in (a) since their oxide-sum were below the threshold. Download button is highlighted by red rectangle.

Downloading the result: The results (in CSV format) can be downloaded using download button near the top-right corner of the “Data table” (highlighted by red rectangle in Figure 5b). The output will appear as given in Figure 5.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
1	Sample	SiO2	MgO	FeO	TiO2	Al2O3	MnO	CaO	Na2O	K2O	Total	Mineral	Confidence	Oxygen_no	Si	Mg	Fe	Ti	Al	Mn	Ca	Na	K	Cation_Total
2	1	50.20	12.80	0.84	0.00	33.50	0.06	0.23	0.26	0.14	98.03	Crd	100	18	5.015	1.910	0.072	0.000	3.952	0.006	0.024	0.048	0.012	11.039
3	2	47.69	7.56	8.04	0.00	32.52	0.04	0.52	0.53	0.42	97.32	Crd	100	18	4.983	1.180	0.703	0.000	4.004	0.006	0.056	0.112	0.050	11.094
4	3	44.64	3.08	11.02	0.00	29.96	1.86	0.02	2.05	0.75	93.38	??	100	0	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
5	4	41.33	19.60	10.44	0.28	21.83	0.44	4.40	0.00	0.00	98.32	Grt	100	12	3.012	2.128	0.635	0.018	1.874	0.026	0.341	0.000	0.000	8.034
6	5	36.70	0.90	29.90	0.75	21.40	1.14	9.02	0.00	0.00	99.81	Grt	100	12	2.950	0.106	2.009	0.044	2.028	0.077	0.777	0.000	0.000	7.991
7	6	39.04	0.73	5.00	0.12	20.43	0.34	34.29	0.00	0.00	99.95	Grt	100	12	2.990	0.083	0.322	0.009	1.841	0.023	2.811	0.000	0.000	8.079
8	7	45.24	0.08	0.02	0.01	36.85	0.12	0.00	0.64	10.08	93.04	??	100	0	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
9	8	45.45	0.69	1.00	0.48	35.99	0.00	0.00	0.65	10.79	95.05	Ms	100	11	3.036	0.068	0.056	0.024	2.835	0.000	0.000	0.080	0.924	7.023
10	9	50.72	2.37	3.70	0.49	27.65	0.03	0.00	0.17	11.53	96.66	Ms	100	11	3.378	0.236	0.204	0.024	2.169	0.000	0.000	0.024	0.976	7.011
11	10	27.22	2.34	13.78	0.56	54.16	0.23	0.00	0.00	0.00	98.29	St	100	46	7.534	0.965	3.193	0.116	17.662	0.050	0.000	0.000	0.000	29.520
12	11	26.62	5.31	9.75	1.67	55.15	0.03	0.00	0.00	0.00	98.53	St	100	46	7.229	2.154	2.219	0.342	17.656	0.000	0.000	0.000	0.000	29.600
13	12	50.20	12.80	0.84	0.00	33.50	0.06	0.23	0.26	0.14	98.03	Crd	100	18	5.015	1.910	0.072	0.000	3.952	0.006	0.024	0.048	0.012	11.039
14	13	47.69	7.56	8.04	0.00	32.52	0.04	0.52	0.53	0.42	97.32	Crd	100	18	4.983	1.180	0.703	0.000	4.004	0.006	0.056	0.112	0.050	11.094
15	14	44.64	3.08	11.02	0.00	29.96	1.86	0.02	2.05	0.75	93.38	??	100	0	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
16	15	41.33	19.60	10.44	0.28	21.83	0.44	4.40	0.00	0.00	98.32	Grt	100	12	3.012	2.128	0.635	0.018	1.874	0.026	0.341	0.000	0.000	8.034
17	16	36.70	0.90	29.90	0.75	21.40	1.14	9.02	0.00	0.00	99.81	Grt	100	12	2.950	0.106	2.009	0.044	2.028	0.077	0.777	0.000	0.000	7.991
18	17	39.04	0.73	5.00	0.12	20.43	0.34	34.29	0.00	0.00	99.95	Grt	100	12	2.990	0.083	0.322	0.009	1.841	0.023	2.811	0.000	0.000	8.079
19	18	45.24	0.08	0.02	0.01	36.85	0.12	0.00	0.64	10.08	93.04	??	100	0	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
20	19	45.45	0.69	1.00	0.48	35.99	0.00	0.00	0.65	10.79	95.05	Ms	100	11	3.036	0.068	0.056	0.024	2.835	0.000	0.000	0.080	0.924	7.023
21	20	50.72	2.37	3.70	0.49	27.65	0.03	0.00	0.17	11.53	96.66	Ms	100	11	3.378	0.236	0.204	0.024	2.169	0.000	0.000	0.024	0.976	7.011
22	21	27.22	2.34	13.78	0.56	54.16	0.23	0.00	0.00	0.00	98.29	St	100	46	7.534	0.965	3.193	0.116	17.662	0.050	0.000	0.000	0.000	29.520
23	22	26.62	5.31	9.75	1.67	55.15	0.03	0.00	0.00	0.00	98.53	St	100	46	7.229	2.154	2.219	0.342	17.656	0.000	0.000	0.000	0.000	29.600
24	23	50.20	12.80	0.84	0.00	33.50	0.06	0.23	0.26	0.14	98.03	Crd	100	18	5.015	1.910	0.072	0.000	3.952	0.006	0.024	0.048	0.012	11.039
25	24	47.69	7.56	8.04	0.00	32.52	0.04	0.52	0.53	0.42	97.32	Crd	100	18	4.983	1.180	0.703	0.000	4.004	0.006	0.056	0.112	0.050	11.094
26	25	44.64	3.08	11.02	0.00	29.96	1.86	0.02	2.05	0.75	93.38	??	100	0	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
27	26	41.33	19.60	10.44	0.28	21.83	0.44	4.40	0.00	0.00	98.32	Grt	100	12	3.012	2.128	0.635	0.018	1.874	0.026	0.341	0.000	0.000	8.034
28	27	36.70	0.90	29.90	0.75	21.40	1.14	9.02	0.00	0.00	99.81	Grt	100	12	2.950	0.106	2.009	0.044	2.028	0.077	0.777	0.000	0.000	7.991
29	28	39.04	0.73	5.00	0.12	20.43	0.34	34.29	0.00	0.00	99.95	Grt	100	12	2.990	0.083	0.322	0.009	1.841	0.023	2.811	0.000	0.000	8.079
30	29	45.24	0.08	0.02	0.01	36.85	0.12	0.00	0.64	10.08	93.04	??	100	0	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
31	30	45.45	0.69	1.00	0.48	35.99	0.00	0.00	0.65	10.79	95.05	Ms	100	11	3.036	0.068	0.056	0.024	2.835	0.000	0.000	0.080	0.924	7.023
32	31	50.72	2.37	3.70	0.49	27.65	0.03	0.00	0.17	11.53	96.66	Ms	100	11	3.378	0.236	0.204	0.024	2.169	0.000	0.000	0.024	0.976	7.011
33	32	27.22	2.34	13.78	0.56	54.16	0.23	0.00	0.00	0.00	98.29	St	100	46	7.534	0.965	3.193	0.116	17.662	0.050	0.000	0.000	0.000	29.520
34	33	26.62	5.31	9.75	1.67	55.15	0.03	0.00	0.00	0.00	98.53	St	100	46	7.229	2.154	2.219	0.342	17.656	0.000	0.000	0.000	0.000	29.600

Figure 5: Screenshot of downloaded result.